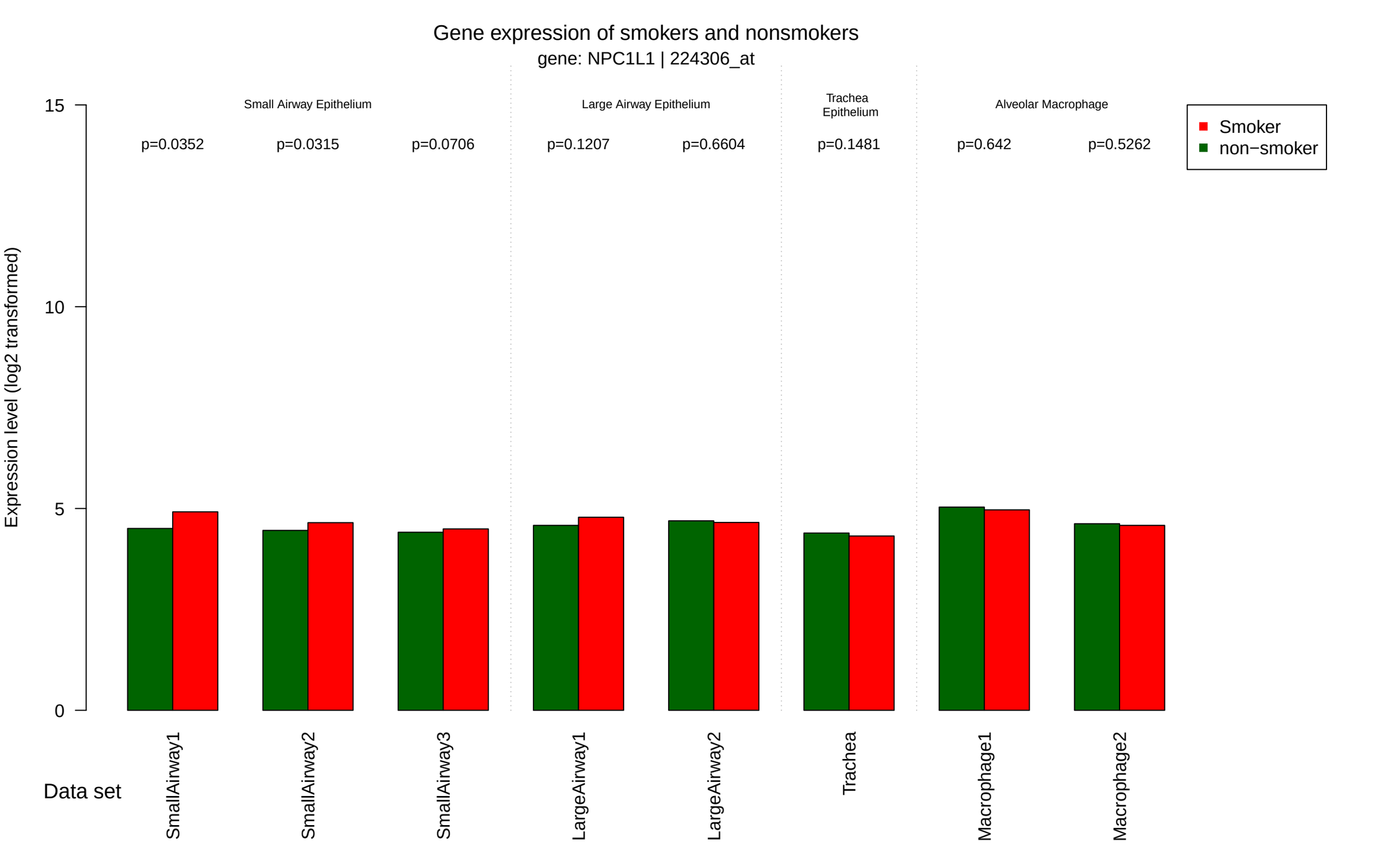
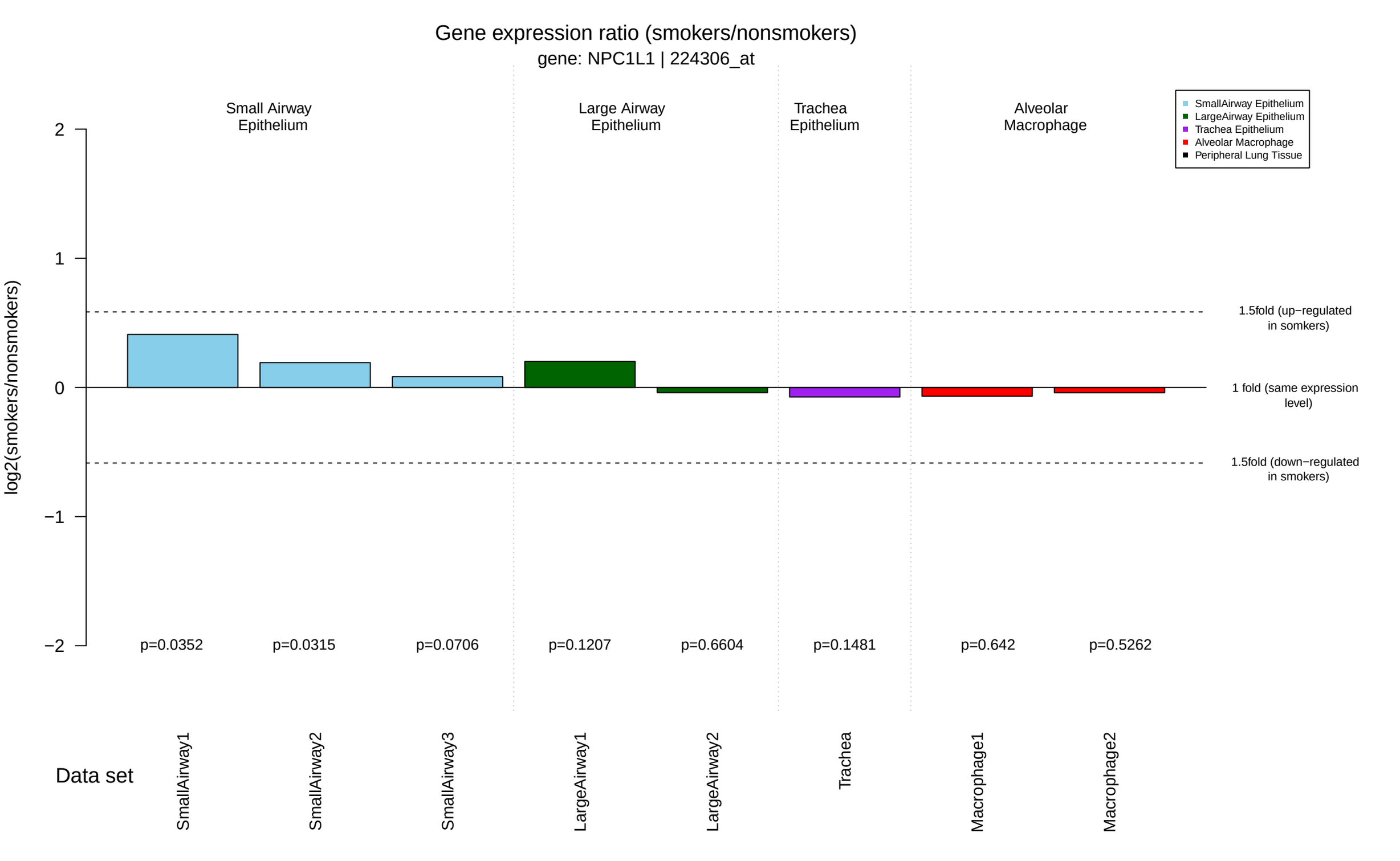
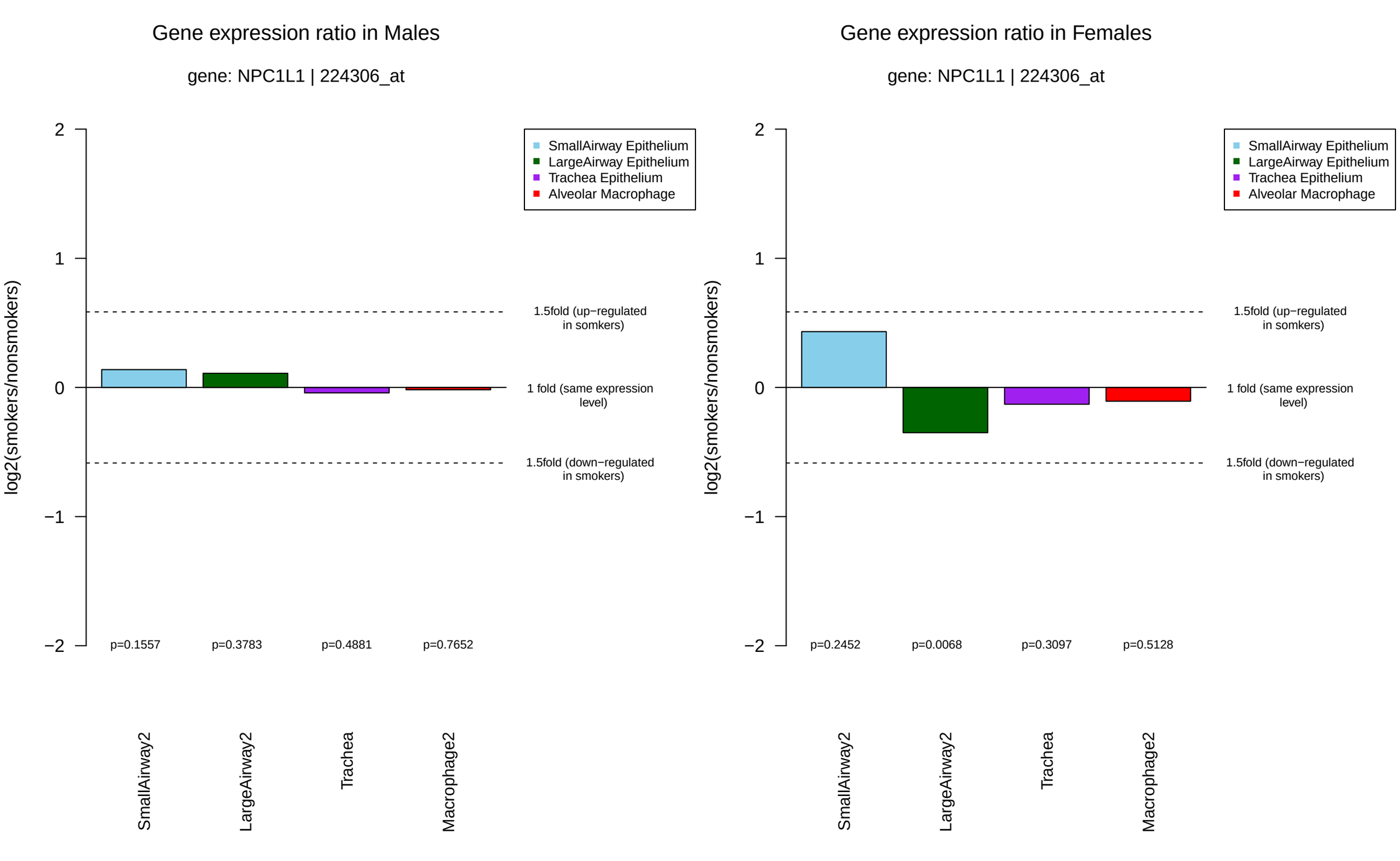
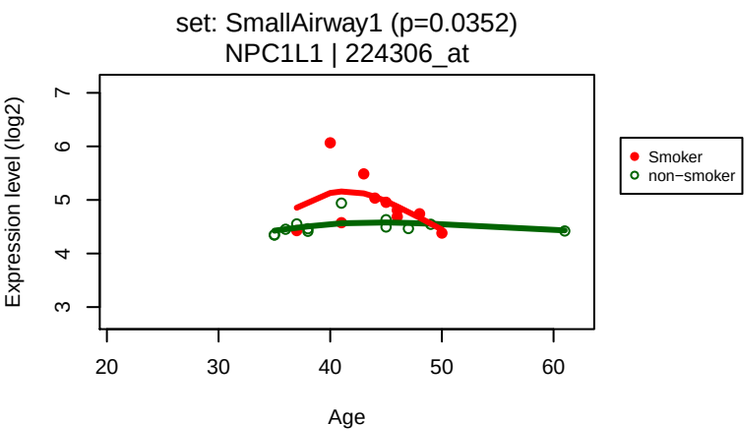
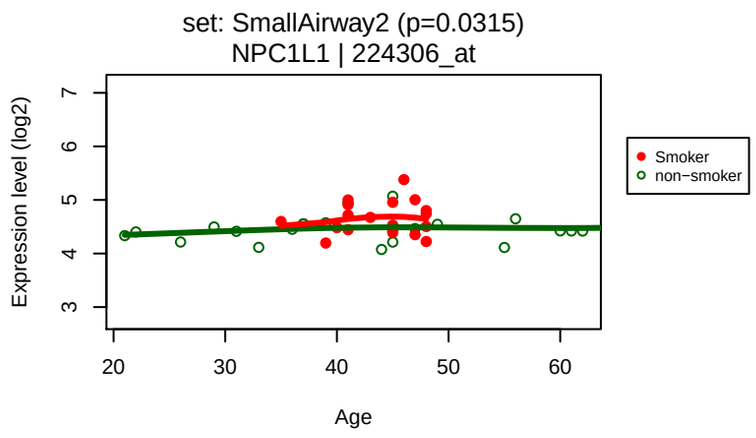
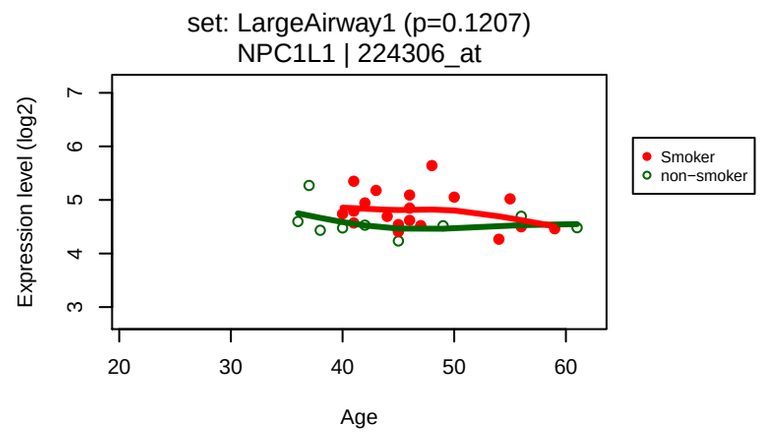

Gene expression of smokers and nonsmokers
gene: NPC1L1 | 224306_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0352
p=0.0315
p=0.0706
p=0.1207
p=0.6604
p=0.1481
p=0.642
p=0.5262
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

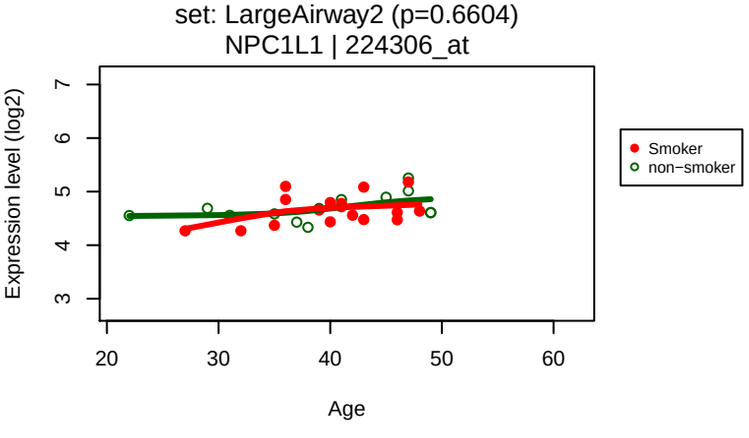

Gene expression ratio (smokers/nonsmokers)
gene: NPC1L1 | 224306_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0352
p=0.0315
p=0.0706
p=0.1207
p=0.6604
p=0.1481
p=0.642
p=0.5262
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

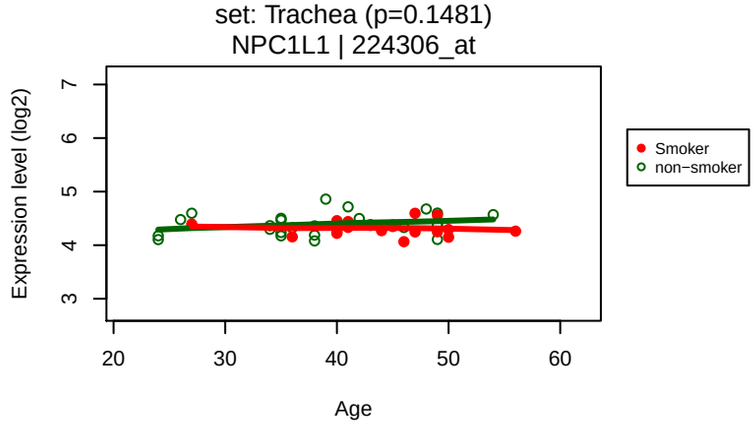

Gene expression ratio in Males
gene: NPC1L1 | 224306_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1557
p=0.3783
p=0.4881
p=0.7652
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NPC1L1 | 224306_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2452
p=0.0068
p=0.3097
p=0.5128
SmallAirway2
LargeAirway2
Trachea
Macrophage2

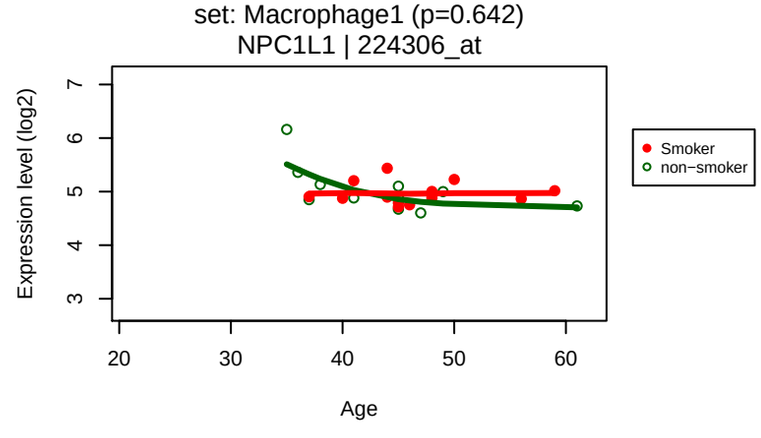

set: SmallAirway1 (p=0.0352)
NPC1L1 | 224306_at
Expression level (log2)
Age
Smoker
non−smoker

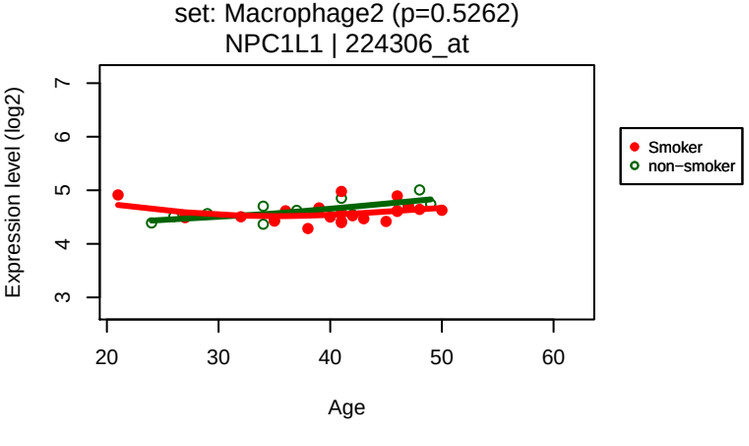

set: SmallAirway2 (p=0.0315)
NPC1L1 | 224306_at
Expression level (log2)
Age
Smoker
non−smoker

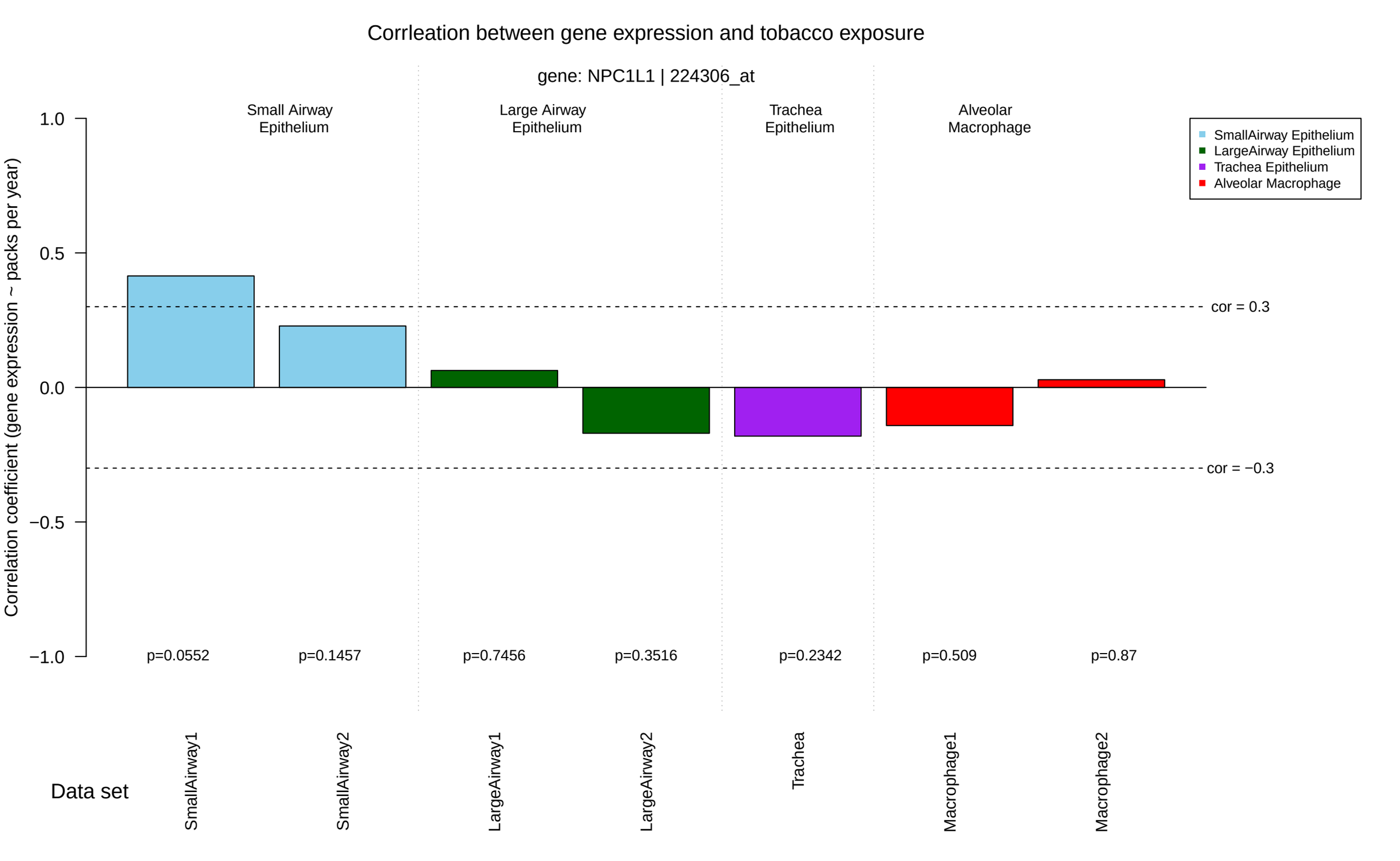

set: LargeAirway1 (p=0.1207)
NPC1L1 | 224306_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6604)
NPC1L1 | 224306_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1481)
NPC1L1 | 224306_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.642)
NPC1L1 | 224306_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5262)
NPC1L1 | 224306_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NPC1L1 | 224306_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0552
p=0.1457
p=0.7456
p=0.3516
p=0.2342
p=0.509
p=0.87
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2